ANGELA ALTAMIRANO

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https://angelaalta.github.io/engineeringportfolio/index.html

EDUCATION

Cornell University, College of Engineering, Ithaca, NY — December 2025
Master of Engineering in Mechanical Engineering, GPA: 4.1

Cornell University, College of Engineering, Ithaca, NY — May 2025
Bachelor of Science in Biomedical Engineering, Cum Laude, GPA: 3.6
Awards: Ryan's Scholar, Louis Stokes Alliance for Minority Participation (LSAMP) Scholar, Dean's List (4 Semesters)

SPECIALIZED SKILLS

Programming Languages: Matlab, Python, RStudio, Julia

Modeling & Simulation: Autodesk Fusion 360, SolidWorks, ANSYS Fluent, COMSOL, AFNI, ImageJ, Microsoft Office Suite

Laboratory & Manufacturing: Cell Culture, Southern Blot Analysis with PCR, Signal Measuring with Oscilloscope, Soldering and Circuit Design, Arduino, 3D Printing, Manual Mill, Manual Lathe, GD&T, DFMA, DFMEA

RELEVANT EXPERIENCE

Master's Thesis: GERDSense, Ithaca, NY, Mechanical Lead — August 2025 - December 2025
- Created and tested over 7 design iterations in Fusion 360 for a pacifier-integrated millifluidic system that non-invasively detects infant gastroesophageal reflux by transporting saliva samples to an external pH sensor
- Collaborated with Pediatric Gastroenterology Physician sponsors at Weill Cornell Medicine to determine engineering design objectives and optimal clinical outcomes

Senior Capstone Project: EasiStitch, Ithaca, NY, Mechanical Lead — August 2024 - May 2025
- Designed and built a gear-based mechanical system for a 360° rotating curved needle, automating the time-intensive process of suturing and producing SolidWorks models, photorealistic renders, and a functional "works-like" prototype
- Translated cross-disciplinary stakeholder input from emergency, surgical, and pediatric providers in varied economic settings into design adaptations that enhanced device functionality, ergonomics, and usability

Jacobs Institute, Buffalo, NY, Engineering Intern — June - August 2024
- Worked with the cardiovascular modeling R&D team and shadowed operating room clinicians to optimize model haptics and design, testing three different coatings to determine which mimics the lubricity of in vivo vessels the best
- Conducted biomechanical characterization of vascular models under simulated use conditions, performing tensile testing on SolidWorks-designed samples using an Instron-68SC to assess model robustness

Cornell DEBUT, Ithaca, NY, Project Manager — August 2023 - May 2024
- Invented and led a team of 7 R&D Analysts to develop *SteadyStride: A Self-Stabilizing Cane for Parkinson's Disease*, integrating 3D-printed components designed in SolidWorks, MATLAB-based modeling and simulation of device dynamics, and iterative prototyping to optimize tremor mitigation
- Served as Principal Investigator for an IRB-approved human subject study at SUNY Cortland, overseeing experimental protocol, accelerometer-based motion data acquisition, and statistical analysis to quantify device performance
- Awarded First Place at Medtronic/BMES Student Design Competition (October 2024)

Weill Cornell Biomedical Engineering Immersion Program, New York, NY, Scholar — January 2024
- Participated in a two-week clinical immersion program with rotations across Emergency Medicine, Hospital Medicine, Neurosurgery, Anesthesiology, Plastic Surgery, and Radiology, observing workflows and clinical decision-making

ADDITIONAL EXPERIENCE

Cornell Affect and Cognition Lab, Ithaca, NY, Research Assistant — August 2023 - May 2025
- Facilitated MRI scans for Parkinson's Disease patients and utilized AFNI for image analysis

Cornell Engineering: Biomedical Circuits, Signals and Systems, Ithaca, NY, Tutor — August - December 2024
- Provided individualized tutoring for BME 3030, supporting student mastery of topics such as circuit analysis, signal processing, and system modeling in biomedical applications